# Сетевые технологии

## Адресация IPv4 и IPv6. Двойной стек

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# Цель работы

# Основная цель

Изучить методы распределения IPv4/IPv6-адресов, разбиение сетей на подсети и настройку двойного стека в виртуальной лабораторной среде.

# Настройка двойного стека адресации IPv4 и IPv6

# Топология стенда

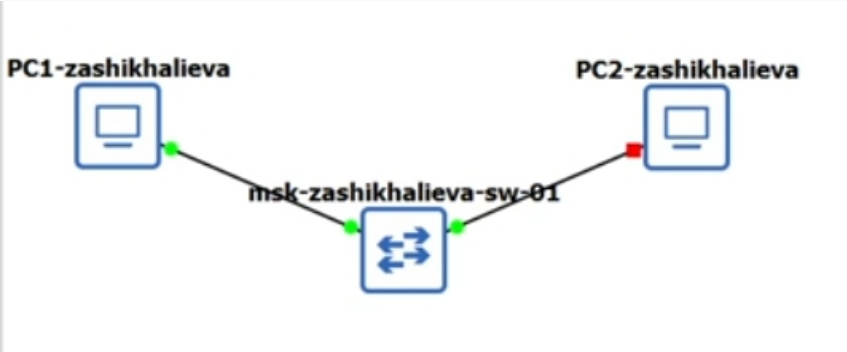



**Рис. 1:** Топология

# Конфигурация PC1

IPv4: 172.16.20.x

IPv6: автоматическое назначение SLAAC

Проверка командой `show ip`

```
Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ?

?                         Print help
arp                         Shortcut for: show arp. Show arp table
clear ARG                 Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION]             Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect                Exit the telnet session (daemon mode)
echo TEXT                 Display TEXT in output. See also  set echo ?
help                      Print help
history                   Shortcut for: show history. List the command history
ip ARG ... [OPTION]       Configure the current VPC's IP settings. See ip ?
load [FILENAME]           Load the configuration/script from the file FILENAME
ping HOST [OPTION ...]    Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit                      Quit program
relay ARG ...             Configure packet relay between UDP ports. See relay ?
rlogin [ip] port          Telnet to port on host at ip (relative to host PC)
save [FILENAME]           Save the configuration to the file FILENAME
set ARG                   Set VPC name and other options. Try set ?
```

# Конфигурация PC2

IPv4: из диапазона 172.16.20.128/26
IPv6: SLAAC

PC2

**Рис. 3:** PC2

# Интерфейсы FRR

- eth0 — 172.16.20.1/25
- eth1 — 172.16.20.129/25
- eth2 — 64.100.1.1/24

Примените фильтр отображения ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | :: | ff02::2 | ICMPv6 | 62 | Router Solicitation |
| 2 | 0.000974 | :: | ff02::2 | ICMPv6 | 62 | Router Solicitation |
| 3 | 0.050587 | Private_66:68:01 | Broadcast | ARP | 64 | Gratuitous ARP for 192.168.1.12 (Request) |
| 4 | 0.051597 | Private_66:68:00 | Broadcast | ARP | 64 | Gratuitous ARP for 192.168.1.11 (Request) |
| 5 | 1.051931 | Private_66:68:01 | Broadcast | ARP | 64 | Gratuitous ARP for 192.168.1.12 (Request) |
| 6 | 1.052757 | Private_66:68:00 | Broadcast | ARP | 64 | Gratuitous ARP for 192.168.1.11 (Request) |

```
> Frame 4: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v Address Resolution Protocol (request/gratuitous ARP)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    [Is gratuitous: True]
    Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
    Sender IP address: 192.168.1.11
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.1.11
```

**Рис. 4:** Настройка интерфейсов

# Проверка конфигурации

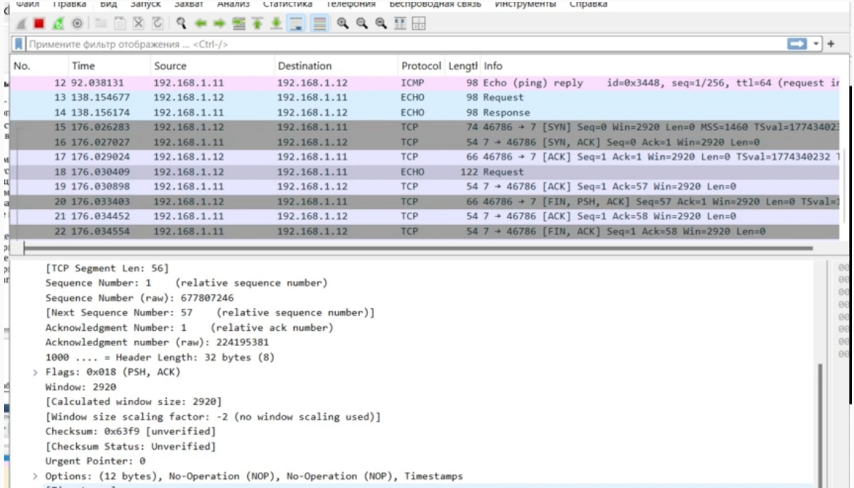
`show running-config` и `show interface brief`

PC2-zashikhalieva - PuTTY

```
    -2              UDP mode
    -3              TCP mode
    -c count        Packet count, default 5
    -D              Set the Don't Fragment bit
    -f FLAG         Tcp header FLAG |C|E|U|A|P|R|S|F|
                                    bits |7 6 5 4 3 2 1 0|
    -i ms           Wait ms milliseconds between sending each packet
    -l size         Data size
    -P protocol     Use IP protocol in ping packets
                      1 - ICMP (default), 17 - UDP, 6 - TCP
    -p port         Destination port
    -s port         Source port
    -T ttl          Set ttl, default 64
    -t              Send packets until interrupted by Ctrl+C
    -w ms           Wait ms milliseconds to receive the response

 Notes: 1. Using names requires DNS to be set.
        2. Use Ctrl+C to stop the command.

PC5> ping 192.168.1.11 -l -c 1

4 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.651 ms

PC5>
```

# Ping и трассировка

- PC1 → PC2
- PC1 → Server




| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 12 | 92.038131 | 192.168.1.11 | 192.168.1.12 | ICMP | 98 | Echo (ping) reply id=0x3448, seq=1/256, ttl=64 (request i |
| 13 | 138.154677 | 192.168.1.12 | 192.168.1.11 | ECHO | 98 | Request |
| 14 | 138.156174 | 192.168.1.11 | 192.168.1.12 | ECHO | 98 | Response |
| 15 | 176.026283 | 192.168.1.12 | 192.168.1.11 | TCP | 74 | 46786 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=17743402 |
| 16 | 176.027027 | 192.168.1.11 | 192.168.1.12 | TCP | 54 | 7 → 46786 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0 |
| 17 | 176.029024 | 192.168.1.12 | 192.168.1.11 | TCP | 66 | 46786 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1774340232 |
| 18 | 176.030409 | 192.168.1.12 | 192.168.1.11 | ECHO | 122 | Request |
| 19 | 176.030898 | 192.168.1.11 | 192.168.1.12 | TCP | 54 | 7 → 46786 [ACK] Seq=1 Ack=57 Win=2920 Len=0 |
| 20 | 176.033403 | 192.168.1.12 | 192.168.1.11 | TCP | 66 | 46786 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval= |
| 21 | 176.034452 | 192.168.1.11 | 192.168.1.12 | TCP | 54 | 7 → 46786 [ACK] Seq=1 Ack=58 Win=2920 Len=0 |
| 22 | 176.034554 | 192.168.1.11 | 192.168.1.12 | TCP | 54 | 7 → 46786 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0 |

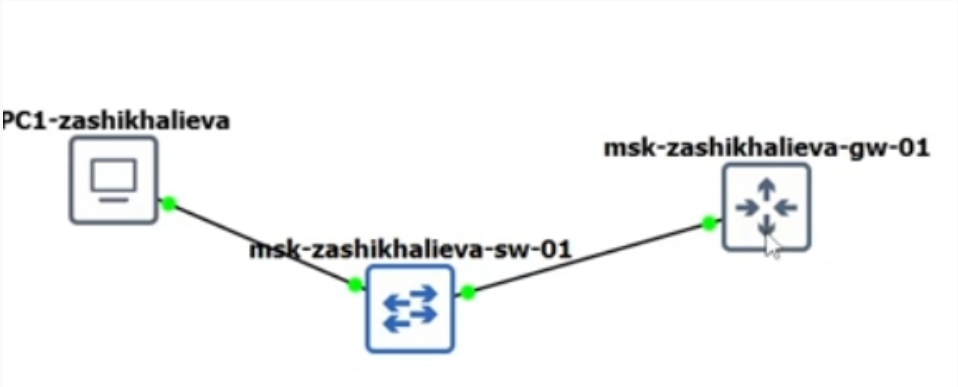
```
[TCP Segment Len: 56]
Sequence Number: 1    (relative sequence number)
Sequence Number (raw): 677807246
[Next Sequence Number: 57    (relative sequence number)]
Acknowledgment Number: 1    (relative ack number)
Acknowledgment number (raw): 224195381
1000 .... = Header Length: 32 bytes (8)
> Flags: 0x018 (PSH, ACK)
Window: 2920
[Calculated window size: 2920]
[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0x63f9 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

# Анализ трафика

ARP-запросы и ICMP-ответы

ARP

**Рис. 7:** ARP



# PC3

Адрес: 2001:db8:c0de:12::/64

```
VPCS> ]
Bad command: "]". Use ? for help.

VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10004
RHOST:PORT  : 127.0.0.1:10005
MTU         : 1500

VPCS>
```

# PC4

Адрес: 2001:db8:c0de:13::/64

```
Hello, this is FRRouting (version 8.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-zashikhalieva-gw-01
msk-zashikhalieva-gw-01(config)# exit
msk-zashikhalieva-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-zashikhalieva-gw-01#
msk-zashikhalieva-gw-01# configure terminal
msk-zashikhalieva-gw-01(config)# interface eth0
msk-zashikhalieva-gw-01(config-if)# ip address 192.168.1.1/24
msk-zashikhalieva-gw-01(config-if)# no shutdown
msk-zashikhalieva-gw-01(config-if)# exit
msk-zashikhalieva-gw-01(config)# exit
msk-zashikhalieva-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
```

# Сервер

IPv6: 2001:db8:c0de:11::/64

Двойной стек: IPv4 + IPv6

```
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-zashikhalieva-gw-01#
msk-zashikhalieva-gw-01# show interface brief
Interface       Status  VRF             Addresses
---------       ------  ---             ---------
eth0            up      default         192.168.1.1/24
eth1            down    default
eth2            down    default
eth3            down    default
eth4            down    default
eth5            down    default
eth6            down    default
eth7            down    default
lo              up      default
pimreg          up      default

msk-zashikhalieva-gw-01#
```

# Назначение адресов и RA

```
VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=2.786 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=3.089 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=4.914 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=3.560 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=2.255 ms
```

**Рис. 12:** VyOS IPv6

# Проверка интерфейсов

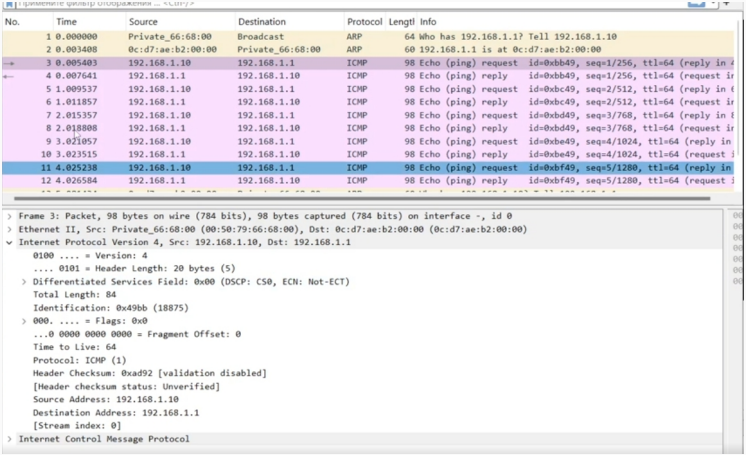



**Рис. 13:** Interfaces VyOS

# Ping PC4 → PC3 и Server

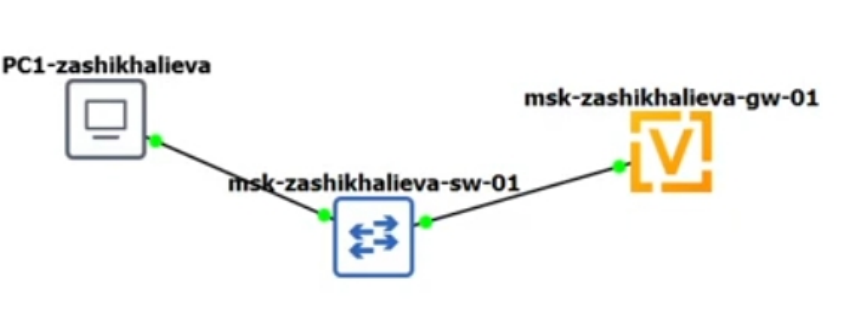



**Рис. 14:** Ping PC4

# ICMPv6-трафик

```
[edit system]
>host-name msk-zashikhalieva-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address 192.168.1.1/24
     hw-id 0c:53:ea:30:00:00
 }
 ethernet eth1 {
     hw-id 0c:53:ea:30:00:01
 }
 ethernet eth2 {
     hw-id 0c:53:ea:30:00:02
 }
 loopback lo {
 }
[edit]
vyos@vyos#
```

# Итоги работы

# Основные результаты

- Выполнено детальное разбиение IPv4/IPv6-сетей
- Настроены двойные стеки на ПК и маршрутизаторах
- Проверена маршрутизация, ARP/ND, ICMPv4/ICMPv6
- Подтверждена корректная работа сетей и взаимодействие подсетей только через маршрутизатр